

Real-Time Transaction Fraud Decision Engine: Video Walkthrough Script

This script is engineered for a high-energy, 90 to 120-second Loom screen recording.

Core Principle: Every spoken sentence directly guides the viewer's eyes to an active visual element, animation, or click interaction happening on screen. Zero abstract lectures - 100% on-screen demonstration.

Recording Setup (Windows)

- Tool: Loom Desktop App (or native Windows Game Bar `Win + G` / OBS Studio).
- Webcam Placement: Bottom-left circular bubble (leaves top KPIs, stream feed, and right-hand forensics panel unobstructed).
- Screen Resolution: 1080p (1920x1080) at 100% or 125% Windows display scaling.
- Browser Window: Chrome maximized (press `F11` for clean presentation with clean navigation).
- Starting Tab: Dashboard (`Tab 1` in sidebar).
- Live Cloud Run URL: `https://fraud-decision-engine-486147352632.us-central1.run.app`

Storyboard Timeline Overview

- 0:00 - 0:25 | Act 1: Live Stream Ticker & Financial ROI (Dashboard)
 - 0:25 - 0:50 | Act 2: Transaction Forensics, TreeSHAP & Analyst Feedback (Split-Pane)
 - 0:50 - 1:15 | Act 3: Value-Adaptive 3DS Simulator & Attack Injection (Simulator)
 - 1:15 - 1:40 | Act 4: Real-Time Wasserstein Drift & Bot Wave Simulation (Stability)
 - 1:40 - 2:00 | Act 5: Sub-25ms Engine SLA & Open-Source Outro (Sidebar Cockpit)
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Second-by-Second Script & Stage Directions

Act 1: Live Stream Ticker & Financial ROI (0:00 – 0:25)

VISUAL TARGET Dashboard (**Tab 1** in sidebar).

- Move cursor across the live streaming table. Point to the rolling status badges: **APPROVE** (Green), **3DS STEP-UP** (Amber), and **DECLINE** (Rose).
- Glance up at the 4 top KPI cards: **Holdout Stream Ingest**, **Direct Approvals**, **Dynamic 3DS Challenges**, and **Portfolio Chargeback Rate (0.40%)**.
- Move cursor to the large Financial ROI card directly below: **71.8% Net Loss Reduction (+\$243k Net Cash Saved)**.

SPOKEN SCRIPT (VERBATIM)

"What you are looking at is a live, production fraud decision engine scoring credit card transactions in real time on Google Cloud Run.

Notice the live feed: transactions aren't simply approved or rejected. They're routed through a dynamic three-way decision system - instant frictionless approvals, low-cost EMV 3DS challenges, and hard declines.

Up top, our portfolio chargeback rate holds at 0.40%, well under Visa and Mastercard's 1.0% regulatory fine thresholds. And right below, our value-adaptive router has slashed total financial fraud loss by 71.8%, preserving over \$243,000 in EBITDA compared to standard static models."

Act 2: Transaction Forensics, TreeSHAP & Analyst Feedback (0:25 – 0:50)

VISUAL TARGET

1. Click on any streaming row with an amber '**3DS STEP-UP**' or rose '**DECLINE**' badge.
2. Watch the right-hand panel ('**Selected Transaction Forensics & Actions**') immediately update with a focus highlight.
3. Point to the '**Primary Risk Attribution (TreeSHAP)**' card showing the top 3 human-readable drivers (e.g. Phone Burst Velocity, Purchaser Email Risk).
4. Scroll down in the right panel and click the '**Flag Dispute**' button.
5. Point to the green confirmation toast: '**Flagged as Chargeback • Written to SQLite**'.

SPOKEN SCRIPT (VERBATIM)

"Let's click on this challenged transaction.

Immediately on the right, you get complete forensics and real-time TreeSHAP explainability: here, high phone burst velocity and email anomaly drove risk up to 24%. Clean approvals bypass TreeSHAP in under 2ms on our fast-path, while flagged payments receive instant C++ attribution in 12ms.

And down here is our analyst station. When an investigator verifies fraud, clicking 'Flag Dispute' logs the ground-truth chargeback label immediately into our operational feedback store."

Act 3: Value-Adaptive 3DS Simulator & Attack Injection (0:50 – 1:15)

VISUAL TARGET

1. Click on '**3DS Simulator**' (**Tab 2** in the sidebar).
2. Point to the continuous **Bayesian Policy Spectrum** belt with the green Approval zone, amber 3DS Step-Up zone, and rose Decline zone.
3. Click the '**High-Velocity Burst**' preset card (showing 14 tx / 5m, \$350.00).
4. Watch the animated probability pin slide dynamically into the rose zone, updating the decision box to **DECLINE**.

SPOKEN SCRIPT (VERBATIM)

"Switching to the 3DS Simulator, we can interactively test payment scenarios against our Bayesian policy spectrum.

The thresholds adapt dynamically to transaction size: high-ticket purchases tighten the approval boundary to protect merchant capital, while low-ticket orders expand it to eliminate cart friction.

Watch what happens when I inject a high-velocity burst attack [CLICK]: the risk score jumps to 77%, and the probability pin shifts dynamically past the threshold into a hard policy decline."

Act 4: Real-Time Wasserstein Drift & Bot Wave Simulation (1:15 – 1:40)

VISUAL TARGET

1. Click on '**Drift & Stability**' (**Tab 3** in the sidebar).
2. Point to the right-hand column: '**Dispute Feedback Queue**', showing the dispute count we logged in Act 2.
3. Point to the left-hand **Wasserstein Feature Drift Bar Chart** with the red dashed horizontal line at '**0.100 ALERT LIMIT**'.
4. Point to the bottom pipeline: '**Delayed Feedback Maturity Lifecycle**' (30 to 120-day horizon).
5. Click '**Simulate Drift Wave**': watch the velocity and amount bars surge past 0.10 and turn red with an alert toast. Then click '**Restore Safe Baseline**'.

SPOKEN SCRIPT (VERBATIM)

"In our Drift and Stability Center, our logged dispute is waiting in the queue. But in card networks, ground-truth chargebacks take 30 to 120 days to settle. If you wait for labels, your model fails silently for months.

So this bar chart monitors unsupervised Wasserstein-1 feature drift across amounts, velocity bursts, and score distributions.

The red dashed line is our 0.10 alert ceiling. Watch what happens when a bot attack hits [CLICK Simulate Drift Wave]: our velocity and amount features spike past the limit, triggering an immediate real-time drift alert before chargebacks ever mature."

Act 5: Sub-25ms Engine SLA & Open-Source Outro (1:40 – 2:00)

VISUAL TARGET

- Point cursor to the left sidebar bottom card: **Engine SLA** (`p95 < 25ms` green badge, ~4.5ms avg, ~16ms p95).
- Switch back to '**Dashboard**' (`Tab 1`) to end on the live streaming view.

SPOKEN SCRIPT (VERBATIM)

"Down here in the sidebar telemetry cockpit, our engine sustains an empirical 16-millisecond p95 latency on Google Cloud Run, comfortably beating enterprise 25ms SLAs.

The entire system - including DuckDB feature pipelines, Feast dual-tier hydration, and automated Locust load tests - is fully open source.

The live Cloud Run application is deployed right now. Click the link in the description to test it live. Thanks for watching!"

Pre-Recording Checklist (60 Seconds)

1. **Verify Live Endpoint:** Open `https://fraud-decision-engine-486147352632.us-central1.run.app` in Chrome.
2. **Display Scaling:** Set Chrome zoom to 100% (or press `F11` for full-screen kiosk view).
3. **Verify Stream Active:** Confirm the stream counter on StatCard 1 is ticking and recent transactions are flowing.
4. **Clean Feedback Queue:** If desired, click `Reset Queue` in Tab 3 so you start with 0 disputes.
5. **Webcam Positioning:** Place your webcam circular bubble at the bottom-left corner so all cards, charts, and right-hand drawer remain 100% visible.

Real-Time Transaction Fraud Engine • Live Cloud Run: <https://fraud-decision-engine-486147352632.us-central1.run.app> • Dual-Tier Feast + Bayesian Dynamic Router + Evidently AI